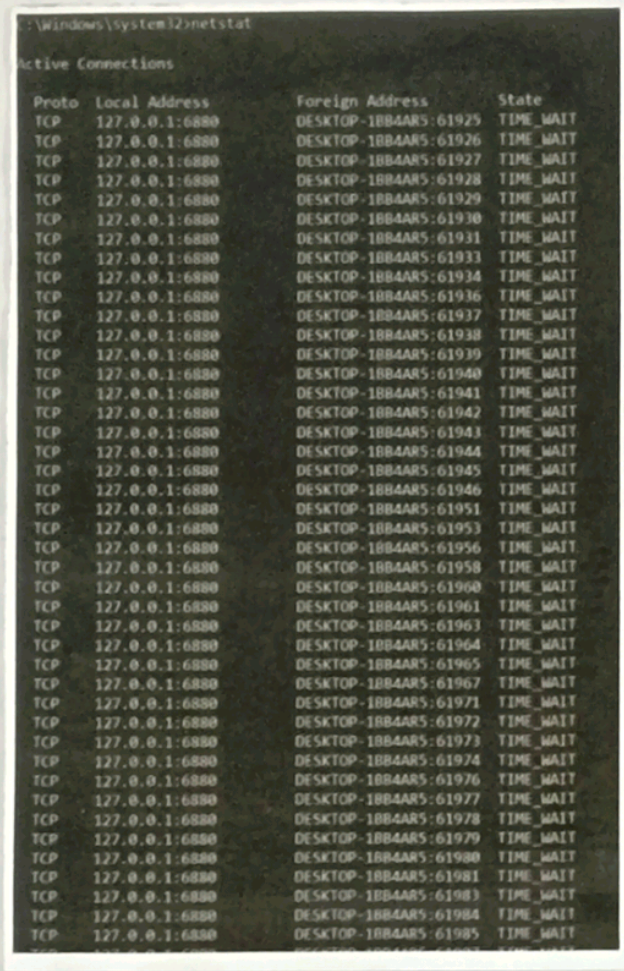
Exp:01 **Basic Networking Commands** 16/07/26

AIM:

To study & execute basic networking commands in windows & Linux.

WINDOWS COMMANDS:

1. ipconfig
   - Displays IP Address. subnet mask, default Gateway.
2. ipconfig /all
   - Shows detailed network information
3. ping google.com
   - Test network Connectivity.
4. tracert google.com
   - Trace Route to Destination
5. nslookup google.com
   - Display DNS information
6. route print
   - Displays routing table
7. netstat
   - Displays active connections
8. netstat -a
   - Display network statistics
9. getmac
   - Display MAC address
10. hostname
    - Check hostname
11. arp -a
    - View ARP table



```
C:\Windows\system32>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61925  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61926  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61927  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61928  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61929  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61930  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61931  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61933  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61934  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61936  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61937  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61938  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61939  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61940  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61941  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61942  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61943  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61944  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61945  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61946  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61951  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61953  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61956  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61958  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61960  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61961  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61963  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61964  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61965  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61967  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61971  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61972  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61973  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61974  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61976  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61977  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61978  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61979  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61980  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61981  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61983  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61984  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:61985  TIME_WAIT
```



```
Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::17d0:9b58:7ce7:6171%15
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

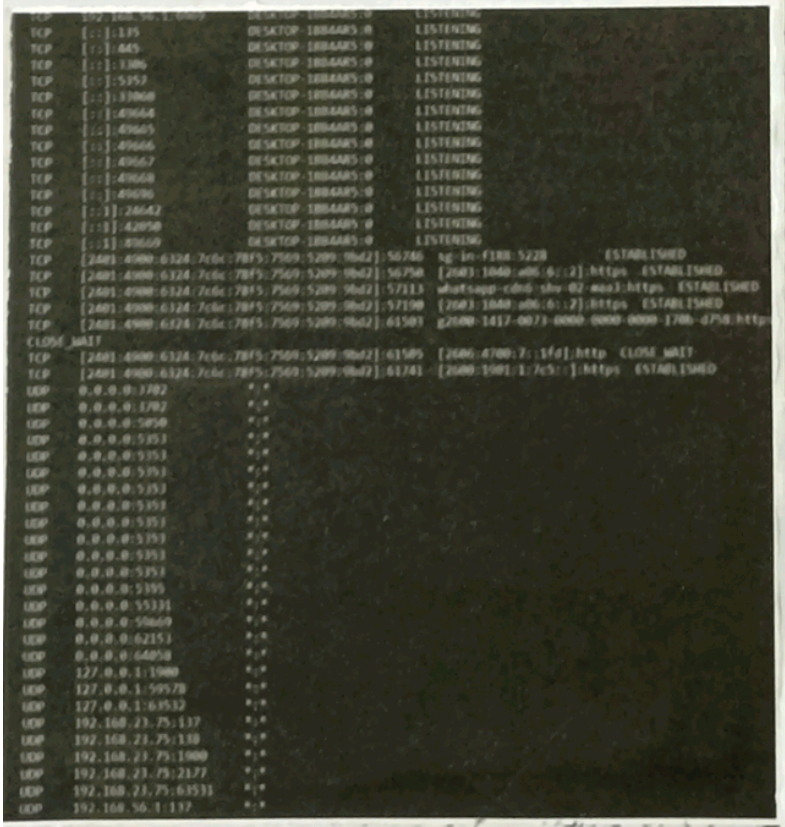
   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:6324:7c6c:498e:1481:6a60:42e2
   Temporary IPv6 Address. . . . . . : 2401:4900:6324:7c6c:78f5:7569:5209:9bd2
   Link-local IPv6 Address . . . . . : fe80::75d1:e75c:bed3:98e5%12
   IPv4 Address. . . . . . . . . . . : 192.168.23.75
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::ccf8:4fff:fe18:e1a7%12
                                       192.168.23.81
```



```
C:\Windows\system32>netstat -a

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:445            DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:2968           DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:3306           DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:5040           DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:5357           DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:33060          DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:49664          DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:49665          DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:49666          DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:49667          DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:49668          DESKTOP-1BB4AR5:0      LISTENING
  TCP    0.0.0.0:49696          DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62112  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62113  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62114  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62116  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62125  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62127  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62128  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62129  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62130  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62132  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62135  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62136  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62138  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62139  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62140  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62142  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62144  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62146  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62147  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62148  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62149  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62150  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62152  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62156  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62157  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62159  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62160  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62167  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62174  TIME_WAIT
```



```
  TCP    [::]:135               DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::]:445               DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::]:3306              DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::]:5357              DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::]:33060             DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::]:49664             DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::]:49665             DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::]:49666             DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::]:49667             DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::]:49668             DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::]:49696             DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::1]:26642            DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::1]:42050            DESKTOP-1BB4AR5:0      LISTENING
  TCP    [::1]:49669            DESKTOP-1BB4AR5:0      LISTENING
  TCP    [2401:4900:6324:7c6c:78f5:7569:5209:9bd2]:56746  sg-in-f188:5228       ESTABLISHED
  TCP    [2401:4900:6324:7c6c:78f5:7569:5209:9bd2]:56750  [2603:1040:a06:6::2]:https  ESTABLISHED
  TCP    [2401:4900:6324:7c6c:78f5:7569:5209:9bd2]:57113  whatsapp-cdn6-shv-02-maa1:https  ESTABLISHED
  TCP    [2401:4900:6324:7c6c:78f5:7569:5209:9bd2]:57190  [2603:1040:a06:6::2]:https  ESTABLISHED
  TCP    [2401:4900:6324:7c6c:78f5:7569:5209:9bd2]:61503  g2600-1417-0073-0000-0000-0000-170b-d750:https
  CLOSE_WAIT
  TCP    [2401:4900:6324:7c6c:78f5:7569:5209:9bd2]:61505  [2606:4700:7::1fd]:http  CLOSE_WAIT
  TCP    [2401:4900:6324:7c6c:78f5:7569:5209:9bd2]:61741  [2600:1901:1:7c5::]:https  ESTABLISHED
  UDP    0.0.0.0:1702           *:*
  UDP    0.0.0.0:1702           *:*
  UDP    0.0.0.0:5050           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5353           *:*
  UDP    0.0.0.0:5355           *:*
  UDP    0.0.0.0:55331          *:*
  UDP    0.0.0.0:59669          *:*
  UDP    0.0.0.0:62151          *:*
  UDP    0.0.0.0:64058          *:*
  UDP    127.0.0.1:1900         *:*
  UDP    127.0.0.1:59578        *:*
  UDP    127.0.0.1:63532        *:*
  UDP    192.168.23.75:137      *:*
  UDP    192.168.23.75:138      *:*
  UDP    192.168.23.75:1900     *:*
  UDP    192.168.23.75:2177     *:*
  UDP    192.168.23.75:63531    *:*
  UDP    192.168.56.1:137       *:*
```



```
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62241  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62242  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1BB4AR5:62243  TIME_WAIT
  TCP    127.0.0.1:15292        DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:15393        DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:16498        DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:27017        DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:45621        DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:49675        DESKTOP-1BB4AR5:49676  ESTABLISHED
  TCP    127.0.0.1:49676        DESKTOP-1BB4AR5:49675  ESTABLISHED
  TCP    127.0.0.1:49677        DESKTOP-1BB4AR5:49678  ESTABLISHED
  TCP    127.0.0.1:49678        DESKTOP-1BB4AR5:49677  ESTABLISHED
  TCP    127.0.0.1:49682        DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:49683        DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:50859        DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:57132        DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:57132        DESKTOP-1BB4AR5:62272  ESTABLISHED
  TCP    127.0.0.1:57133        DESKTOP-1BB4AR5:0      LISTENING
  TCP    127.0.0.1:62111        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62114        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62119        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62119        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62120        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62121        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62122        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62124        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62131        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62133        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62134        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62137        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62139        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62141        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62143        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62145        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62153        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62154        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62155        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62160        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62161        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62163        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62164        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62165        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62166        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62171        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62172        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62173        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62175        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62179        DESKTOP-1BB4AR5:6880   TIME_WAIT
  TCP    127.0.0.1:62180        DESKTOP-1BB4AR5:6880   TIME_WAIT
```

LINUX NETWORKING COMMANDS:

1. ip addr / ifconfig (if installed)
   - Displays IP address
2. ip route
   - Display routing table
3. hostname
   - Displays hostname
4. ping google.com
   - Test Connectivity
5. traceroute google.com /
   sudo apt install traceroute
   - Trace network route
6. nslookup google.com /
   sudo apt install dnsutils
   - DNS lookup
7. netstat -a / ss -a
   - Display Network statistics
8. arp -a / ip neigh
   - Display ARP table
9. ip link / ifconfig / link/ether
   - MAC address
10. cat/etc/resolv.conf
    - DNS configuration
11. ip link show
    - Check interface status
12. ss -tuln
    - Displays listening ports
13. curl google.com / wget google.com
    - Check internet connectivity.





```
C:\Windows\system32>ping google.com

Pinging google.com [2404:6800:4007:83f::200e] with 32 bytes of data:
Reply from 2404:6800:4007:83f::200e: time=16ms
Reply from 2404:6800:4007:83f::200e: time=21ms
Reply from 2404:6800:4007:83f::200e: time=29ms
Reply from 2404:6800:4007:83f::200e: time=23ms

Ping statistics for 2404:6800:4007:83f::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 29ms, Average = 22ms
```

```
  UDP    [2401:4900:6324:7c6c:49be:1481:6a6b:42e2]:2177  *:*
  UDP    [2401:4900:6324:7c6c:78f5:7569:5209:9bd2]:2177  *:*
  UDP    [fe80::17d6:8b58:7ce7:6171%15]:1900  *:*
  UDP    [fe80::17d6:8b58:7ce7:6171%15]:2177  *:*
  UDP    [fe80::17d6:8b58:7ce7:6171%15]:63527  *:*
  UDP    [fe80::75d1:e75c:be43:96e5%12]:1900  *:*
  UDP    [fe80::75d1:e75c:be43:96e5%12]:2177  *:*
  UDP    [fe80::75d1:e75c:be43:96e5%12]:63528  *:*

C:\Windows\system32>getmac

Physical Address    Transport Name
=================== ==========================================================
A8-7E-EA-9B-20-80   \Device\Tcpip_{A8B673E4-C372-467A-8DF2-CE891803184F}
98-E7-43-55-23-22   Media disconnected
0A-00-27-00-00-0F   \Device\Tcpip_{B3C18887-3EC7-468D-9F2E-7BFFF9A7AF89}
```

```
C:\Windows\system32>tracert google.com

Tracing route to google.com [2404:6800:4007:83f::200e]
over a maximum of 30 hops:

  1     3 ms     2 ms     5 ms  2401:4900:6324:7c6c::54
  2     *        *        *     Request timed out.
  3    28 ms    15 ms    16 ms  2401:4900:0:7f3::1
  4    45 ms    30 ms    25 ms  2401:4900:0:6fe::2
  5    20 ms    17 ms    22 ms  2401:4900:0:6f7::1
  6     *        *        *     Request timed out.
  7     *       36 ms    19 ms  2404:a800:1a00:1::601
  8    28 ms    12 ms     *     2404:a800::92
  9    16 ms    25 ms    21 ms  2001:4860:1:1::674
 10    48 ms    20 ms    20 ms  2001:4860:0:1::a039
 11    20 ms    21 ms    26 ms  2001:4860:0:1::33cb
 12    42 ms    28 ms    30 ms  prmaaa-at-in-x0e.1e100.net [2404:6800:4007:83f::200e]

Trace complete.
```

```
C:\Windows\system32>nslookup google.com
Server:  UnKnown
Address:  192.168.23.81

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4007:83f::200e
          172.217.24.206
```

```
unix  3      [ ]         STREAM     CONNECTED     30527
unix  3      [ ]         STREAM     CONNECTED     30501
unix  3      [ ]         STREAM     CONNECTED     30401
unix  3      [ ]         STREAM     CONNECTED     30093
unix  3      [ ]         STREAM     CONNECTED     29974
unix  3      [ ]         STREAM     CONNECTED     22327    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     31056    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     30434    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     29349
unix  3      [ ]         STREAM     CONNECTED     31355    /run/dbus/system_bus_
socket
unix  3      [ ]         STREAM     CONNECTED     31066    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     31053
unix  3      [ ]         STREAM     CONNECTED     30426
unix  3      [ ]         STREAM     CONNECTED     30072
unix  3      [ ]         STREAM     CONNECTED     30146    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     31686
Active Bluetooth connections (servers and established)
Proto  Destination       Source            State         PSM DCID   SCID      IM
TU     OMTU Security
Proto  Destination       Source            State         Channel
[dj@localhost ~]$
```

```
[dj@localhost ~]$ nslookup google.com
Server:         192.168.230.203
Address:        192.168.230.203#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.222.174
Name:   google.com
Address: 2404:6800:4007:83b::200e

[dj@localhost ~]$
```

```
[dj@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp6       0      0 localhost:ipp           [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
udp        0      0 localhost.locald:bootpc _gateway:bootps         ESTABLISHED
udp        0      0 0.0.0.0:mdns            0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp6       0      0 [::]:mdns               [::]:*
udp6       0      0 localhost:323           [::]:*
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  2      [ ACC ]     STREAM     LISTENING     33115    /run/user/0/gnupg/d.r
w35a3wcth45h7ype6ibtf71/S.gpg-agent
unix  2      [ ACC ]     STREAM     LISTENING     20832    /run/systemd/private
unix  2      [ ACC ]     STREAM     LISTENING     33116    /run/user/0/gnupg/d.r
w35a3wcth45h7ype6ibtf71/S.gpg-agent.extra
unix  2      [ ACC ]     STREAM     LISTENING     33117    /run/user/0/gnupg/d.r
w35a3wcth45h7ype6ibtf71/S.gpg-agent.browser
unix  2      [ ACC ]     STREAM     LISTENING     33118    /run/user/0/gnupg/d.r
w35a3wcth45h7ype6ibtf71/S.gpg-agent.ssh
```

```
unix  3      [ ]         STREAM     CONNECTED     31720    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     32189    @480bc
unix  3      [ ]         DGRAM      CONNECTED     29346
unix  3      [ ]         STREAM     CONNECTED     32916
unix  3      [ ]         STREAM     CONNECTED     30206    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     30022
unix  3      [ ]         STREAM     CONNECTED     26939
unix  3      [ ]         STREAM     CONNECTED     23768
unix  3      [ ]         STREAM     CONNECTED     32188
unix  3      [ ]         STREAM     CONNECTED     31522
unix  3      [ ]         STREAM     CONNECTED     29431
unix  3      [ ]         STREAM     CONNECTED     30861
unix  3      [ ]         STREAM     CONNECTED     22675
unix  2      [ ]         DGRAM      CONNECTED     22313
unix  3      [ ]         STREAM     CONNECTED     32549    @480bc
unix  3      [ ]         STREAM     CONNECTED     30023    /run/dbus/system_bus_
socket
unix  2      [ ]         DGRAM                    28785
unix  3      [ ]         STREAM     CONNECTED     32470    /run/systemd/journal/
stdout
unix  3      [ ]         STREAM     CONNECTED     30187    /run/user/1000/pipewi
re-0-manager
unix  3      [ ]         STREAM     CONNECTED     29806    /run/systemd/journal/
stdout
```

```
[dj@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[dj@localhost ~]$
```

```
[dj@localhost ~]$ hostname
localhost.localdomain
[dj@localhost ~]$
```

Result:

Hence, the basic computing networks commands for both windows & Linux has been success fully executed & verified.